

HELLYER TECHNOLOGIES

Precision 16

User Guide

Precision Control. Proven Reliability.

Model PREC-P4-16 • 16-zone irrigation controller
Applies to firmware 0.1.0 • July 2026

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1. Introduction & Key Concepts

The Precision 16 is a 16-zone smart irrigation controller with a 7-inch color touch screen. It waters on programmable schedules, adjusts automatically for rain and freezing weather, and can be run entirely by hand when you need it.

Zones

A **zone** is one valve circuit - a sprinkler group covering one area of your landscape. The Precision 16 controls up to 16 zones and always waters them **one at a time**, which protects water pressure and lets every head in a zone perform at its best. Zones are numbered 1 to 16 to match the valve terminals they are wired to, and each can be given a name - see **Zone Names** in section 8.

Schedules

A **schedule** is a named watering program: a start time, the days of the week it runs, and a per-zone run time for each zone it includes. You can store up to **8 schedules**, and each can use any combination of the 16 zones.

The queue

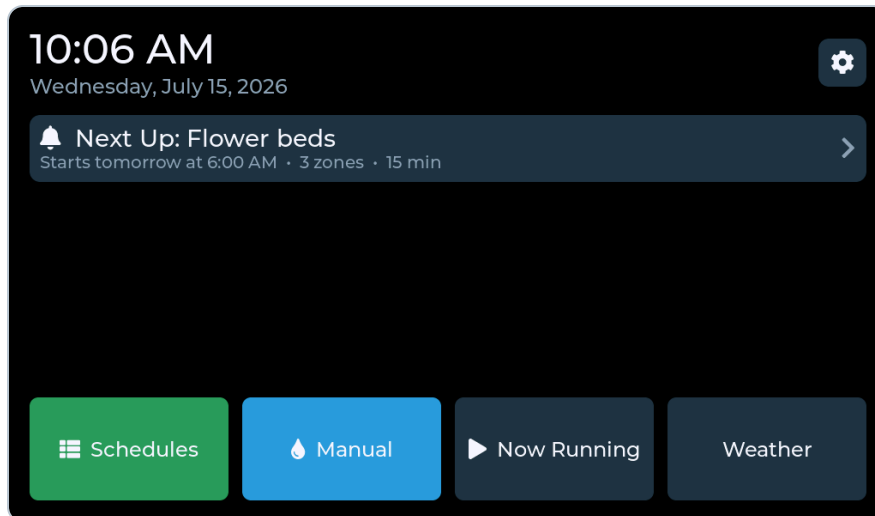
Whenever watering starts - from a schedule or by hand - zones enter a single **run queue**. The first zone runs while the rest wait their turn; between zones the controller pauses for the configurable **zone delay** so valves can close fully. The **Now Running** screen shows and controls this queue live.

Designed to fail open

Weather-based features (rain adjustment, frost hold) only ever act on *current, trusted* data. If Wi-Fi drops, the forecast goes stale, or the clock is not set, these features quietly stand down and your schedules run exactly as programmed. A network outage never silently stops irrigation.

About the screen: The display turns off to save power after sitting idle (touch it to wake). A scheduled start wakes the controller on its own: the display turns on and stays on for as long as watering runs. Sleep timing and brightness are adjustable in Settings.

2. Home Screen

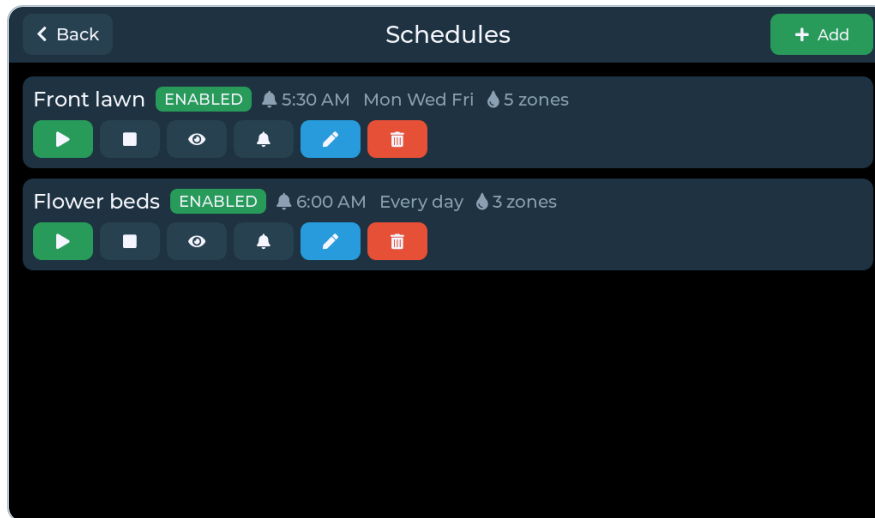


Home is the controller's hub: the current time and date, today's conditions, and one-tap access to every other screen.

What you see

- **Clock and date** - large and readable across the room. Shows "--:--" with "Waiting for network time..." until the clock first syncs.
- **Next Up** - your next scheduled run: schedule name, start ("today at 5:30 AM", "tomorrow..."), zone count, and total duration.
- **Live status card** - appears only while watering: the running zone, time remaining, the source schedule (or "Manual"), and how many zones are queued. Tap it to open Now Running.
- **Notice banners** - contextual cards that appear when they matter:
 - *Set up Wi-Fi* - shown until a network is configured; tap to connect.
 - *Standby Mode* - watering is paused; shows when it resumes. Tap to change.
 - *Frost Hold* - schedules paused because today's forecast low is 40°F or below (see section 12).
 - *Weather note* - "Today's watering was skipped / ran at 70% after recent rain" when Weather Adjustment acted today.
- **Navigation buttons** - **Schedules**, **Manual**, **Now Running**, and **Weather**, plus the gear icon for **Settings**.

3. Schedules



The Schedules screen lists your watering programs. Each row shows the schedule’s name, start time, run days (“Every day”, “Mon Wed Fri”), zone count, and status badge.

Per-schedule actions

The buttons appear on each row in this order, left to right:

Control	What it does
 Play	Starts the whole schedule right now, regardless of its programmed time. A hand-started schedule is treated as manual watering: weather adjustment and frost hold do not touch it.
 Stop	Stops the schedule if it is currently running - its zones leave the queue. The button turns red while the schedule runs.
 Enable	Turns the schedule on or off without deleting it - the badge switches between ENABLED and DISABLED. Disabled schedules keep all their settings.
 Delay	Skips the next day’s automatic start (badge shows “DELAYED 1d”). Tap again to cancel the delay. Handy when rain is on the way or the lawn was just treated.
 Edit	Opens the schedule editor (section 4).
 Delete	Permanently removes the schedule, after a confirmation prompt. This cannot be undone.

Tap **Add** in the header to create a new schedule (up to 8).

4. Schedule Editor

The screenshot shows a mobile app interface for creating a new schedule. The title bar at the top has a back arrow and the word 'Done' in green, followed by the title 'New Schedule'. The main content area has four sections: 1. 'Schedule Name' with a text input field containing 'Front lawn'. 2. 'Schedule Start Time' with a time picker showing 5:30 AM. 3. 'Run on days' with seven day buttons (Sun, Mon, Tue, Wed, Thu, Fri, Sat); Mon, Wed, and Fri are highlighted in green. 4. 'Zones' with a water drop icon, the text '5 zones on - 10-20 min', and a 'Select Zones >' button. A 'Cancel' button is at the bottom.

The editor defines when a schedule runs and what it waters. Changes are held as a draft: nothing takes effect until you tap **Done** in the header, and **Cancel** discards everything - including zone changes made on the zones sub-screen. If anything required is still missing, Done explains what is needed and keeps you in the editor, so a schedule can never be saved half-finished.

Fields

- **Name** - tap to rename in a keyboard dialog. Use names that describe the area (“Front Lawn”, “Drip Beds”).
- **Start time** - hour, minute, and AM/PM rollers.
- **Run days** - seven day buttons, Sunday through Saturday. Any combination is allowed.
- **Zones summary row** - shows how many zones are enabled and the total run time; tap it to open the zones sub-screen.

Zones sub-screen

The screenshot shows a mobile app interface titled "Zones - Front lawn". It features a grid of 16 zones, each with a toggle switch and a minutes stepper. The zones are arranged in two columns of eight. The first column includes "Front lawn", "Side lawn", "Driveway strip", "Front beds", "Front verge", "Flower bed 1", "Flower bed 2", and "Rose garden". The second column includes "Veg patch", "Greenhouse", "Back lawn", "Patio pots", "Herb garden", "Orchard", "Hedge row", and "Compost corner". At the bottom, there is a "Set all zone runtimes" section with a minutes stepper and an "Apply to All" button. A green "Done" button is in the top left corner.

Zone	Toggle	Minutes
Front lawn	On	10 min
Side lawn	On	15 min
Driveway strip	On	10 min
Front beds	On	20 min
Front verge	On	10 min
Flower bed 1	Off	10 min
Flower bed 2	Off	10 min
Rose garden	Off	10 min
Veg patch	Off	10 min
Greenhouse	Off	10 min
Back lawn	Off	10 min
Patio pots	Off	10 min
Herb garden	Off	10 min
Orchard	Off	10 min
Hedge row	Off	10 min
Compost corner	Off	10 min

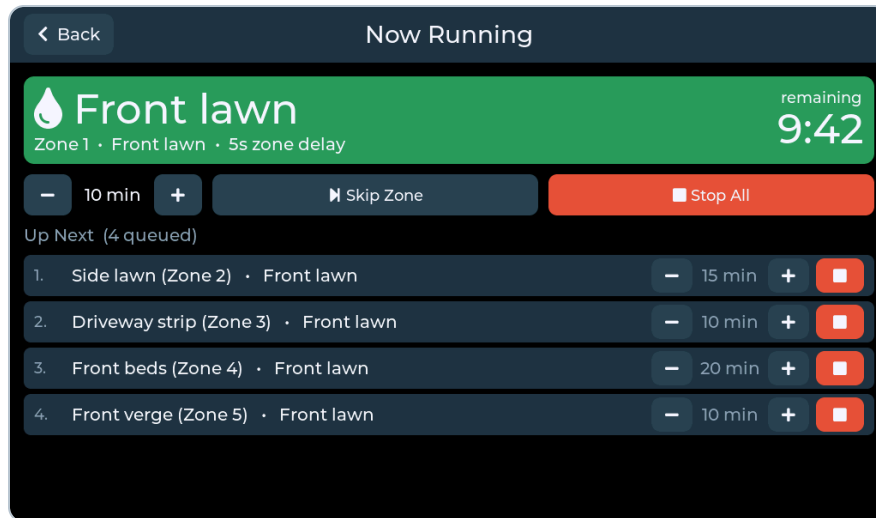
Set all zone runtimes: 10 min [Apply to All]

All 16 zones appear in a fixed grid - nothing scrolls. For each zone:

- The **toggle** includes or excludes the zone from this schedule.
- The **minutes stepper** sets how long that zone runs.

Apply to all zones at the top sets a single run time across every enabled zone in one tap - a fast starting point you can then fine-tune per zone. **Done** returns to the editor; the editor's own **Done** is what finally commits the schedule.

5. Now Running (Watering Queue)



Now Running is the live view of the watering queue - what is watering, what is waiting, and the controls to change either.

The active zone

The top card shows the running zone in large type with its countdown, the schedule it came from (or “Manual”), and the zone delay in effect. While the controller is between zones, the card shows the next zone with “next” during the delay pause.

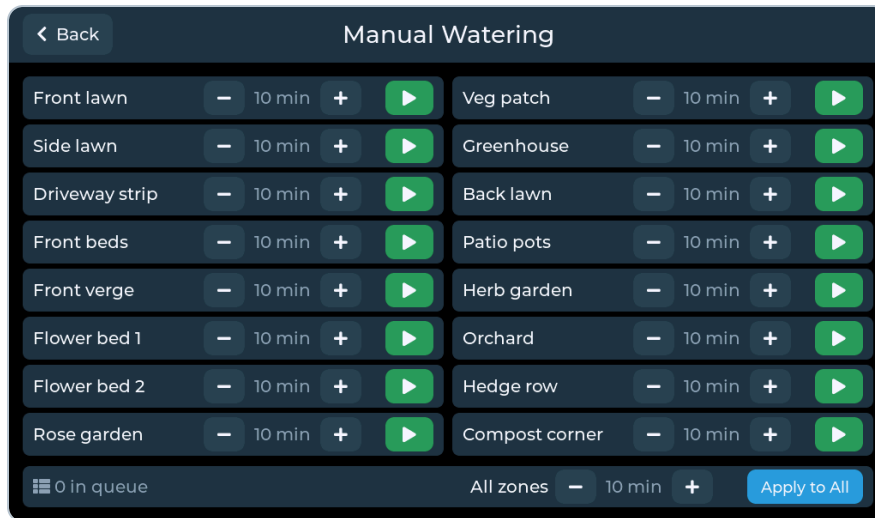
- **Minutes stepper** - extend or trim the running zone’s remaining time on the fly.
- **Stop** - ends the running zone; the next queued zone starts after the zone delay.

Up next

Queued zones are listed in run order (“Up Next (4 queued)”). Each row has its own minutes stepper and a stop button that removes just that entry - the rest of the queue is untouched.

Tip: The queue is the single source of truth for all watering. A schedule start, a manual zone, and a hand-started schedule all flow through the same queue, so this screen always shows the complete picture.

6. Manual Watering



Manual watering runs any zones you choose, right now, without touching your schedules. All 16 zones are shown at once in a fixed grid.

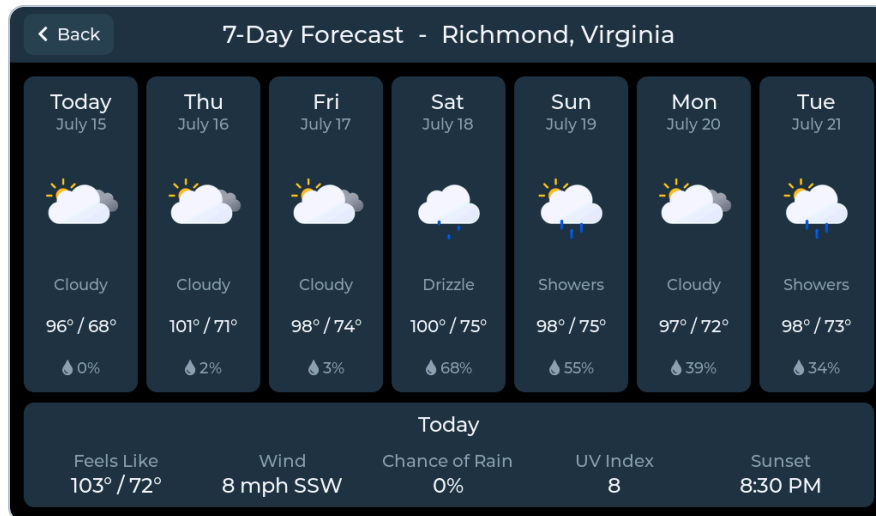
Running zones by hand

1. Set a zone's run time with its - / + stepper (1 to 120 minutes).
2. Tap the zone's **play** button. The first zone starts immediately; additional zones queue behind it and the header shows the queue count.
3. A queued zone's button becomes a **stop** that removes it again.

Apply to all sets one run time on every zone at once, so a full system walk-through (for example, 2 minutes per zone) takes seconds to set up.

Note: Manual watering is deliberately exempt from Weather Adjustment and Frost Hold - when you start a zone by hand, it runs for exactly the time you set. It is blocked only by Standby Mode.

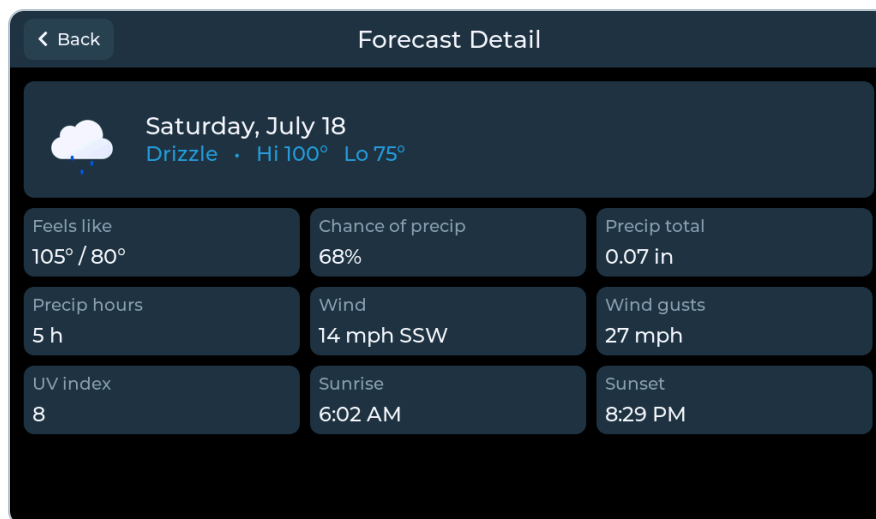
7. Weather & Daily Detail



The Weather screen shows a 7-day local forecast: conditions icon, high/low temperatures, and rain outlook for each day. The forecast refreshes automatically over Wi-Fi throughout the day.

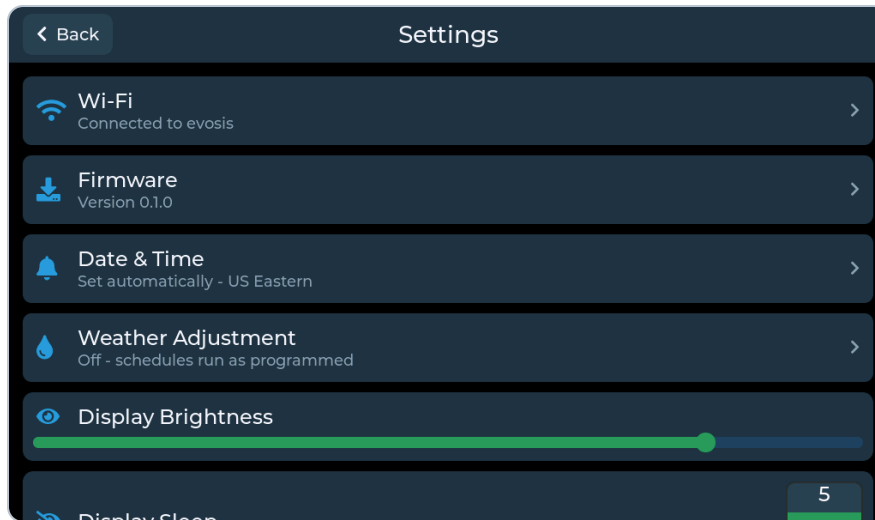
Your forecast location comes from your internet connection, or from the custom location set on the Weather Adjustment screen (section 11).

Daily detail



Tap any day for its detail: conditions summary with high and low, plus a grid of stats including precipitation amount and probability, wind, humidity, UV index, and sunrise/sunset. This is the same data the automatic Weather Adjustment works from, so you can always see the “why” behind a skipped or shortened run.

8. Settings



Settings is the hub for device configuration. The rows at the top open dedicated screens - **Wi-Fi**, **Firmware**, **Date & Time**, **Weather Adjustment**, **Zones**, and **About** - each showing its current status at a glance (for example, “Connected to HomeNet” or “Update available”). The cards below adjust the device directly:

Display Brightness

A live slider for backlight brightness - changes apply as you drag, and the level is remembered across restarts.

Display Sleep

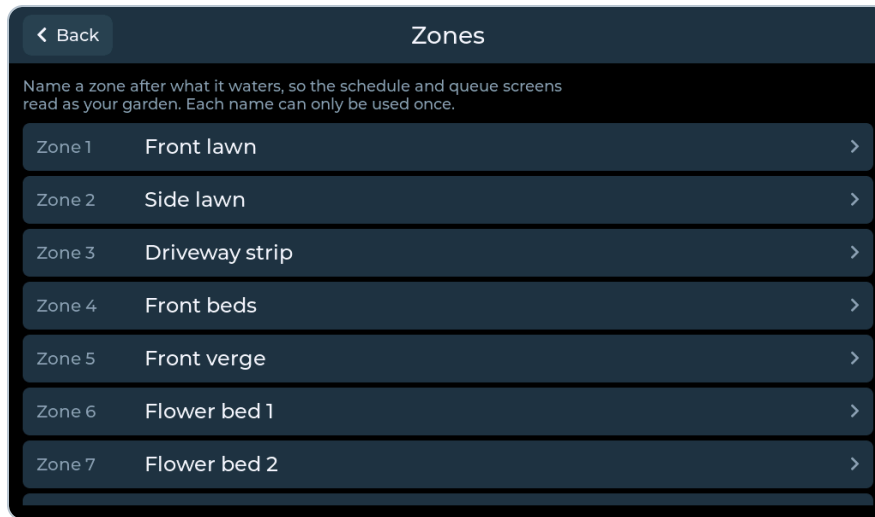
How long the controller waits, idle, before turning the display off to save power: 1 to 60 minutes, or **Never**. Touch the screen to wake it; a scheduled start wakes the controller by itself, and the display stays on while watering runs.

Standby Mode

The master pause switch. While standby is on, all schedules and manual watering are disabled and anything currently running stops immediately.

- Set **days and hours** before switching it on for a timed standby - “Standby on, resumes Sat Jul 12 3:04 PM” - ideal for vacations or lawn treatments.
- Leave both at 0 for an indefinite standby that stays on until you switch it off.
- Home shows a standby banner the whole time, so the paused state is never a mystery.

Zone Names



Each zone can carry a name describing what it waters - “Front lawn”, “Veg patch” - so the schedule, queue and manual screens read as your garden rather than as terminal numbers. Tap **Zones** in Settings, tap any zone in the list, and type its name.

- Names are **optional**, and up to **20 characters**. Clear the field and that zone goes back to plain “Zone 3”.
- Each name can be used **only once**, so a name always points at exactly one zone. Number them where areas repeat - “Flower bed 1”, “Flower bed 2”. If you choose a name that is already taken, the controller tells you which zone holds it and offers the next free number in a single tap.
- The zone number is never lost. Screens with room show both, as “Side lawn (Zone 2)” on the Now Running queue, so a name can always be traced back to the valve terminal it is wired to.

Zone Delay

The minimum pause after one zone stops before the next starts - from 5 seconds up to 10 minutes. Use longer delays if your valves close slowly, or to let a well or low-flow supply recover pressure between zones.

9. Wi-Fi

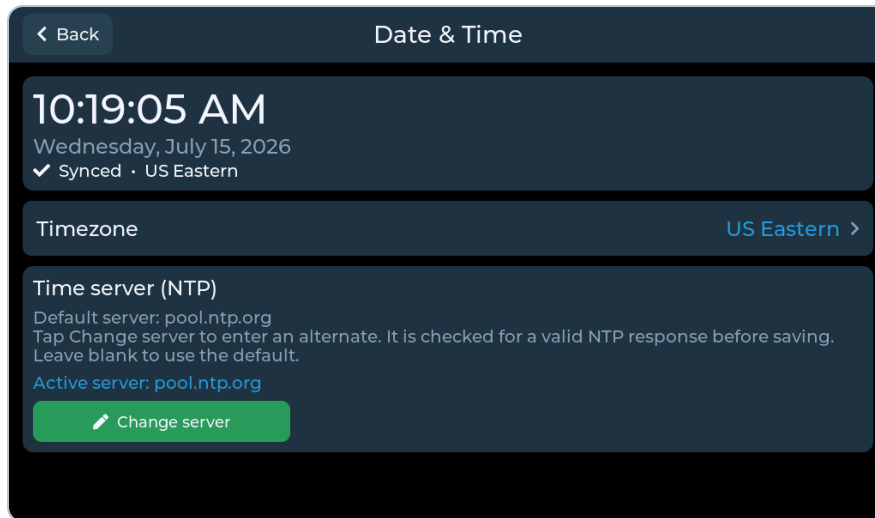


The Wi-Fi screen connects the controller to your home network, which enables the clock, weather forecast, weather-based skipping, and firmware updates. Watering itself never depends on Wi-Fi.

- **Status line** - connected network, connection progress, or what needs attention.
- **Network list** - nearby networks from a live scan; tap yours and enter the password in the keyboard dialog. Passwords are **case sensitive**: upper and lower case letters matter, so type capital letters exactly as they appear in the password.
- The controller stores one network and reconnects automatically at power-up. Connecting to a different network replaces the saved one.

Tip: If the controller sits at “Connection failed”, re-select the network and re-enter the password - a mistyped password is by far the most common cause, and mixed-up upper/lower case letters are the usual culprit.

10. Date & Time

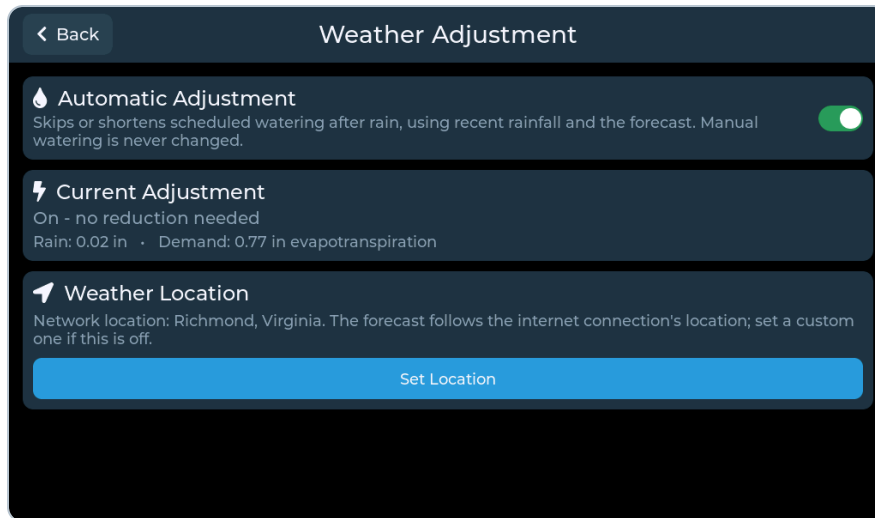


Schedules run on this clock. With Wi-Fi connected the time sets itself from internet time servers and stays accurate indefinitely; your only job is choosing the time zone.

- **Live clock** - with a “Synced” check mark and the active time zone when everything is healthy.
- **Time zone list** - covers U.S. and international zones; daylight saving transitions are automatic. The controller ships set to **US Eastern** - make sure the selection matches your location.
- **Change server** - for networks that block public time servers, a custom NTP server can be entered. The controller validates the server responds before saving it. Leave blank to return to the default.

Heads-up: Until the clock has synced at least once, scheduled starts wait and Home shows “Waiting for network time...”. Manual watering works regardless.

11. Weather Adjustment



Weather Adjustment automatically skips or shortens *scheduled* watering when nature has already done the work. It balances the water supplied - recent rainfall plus today's probability-weighted forecast rain - against the ground's water demand (evapotranspiration), computed for your location.

How it decides

Condition	Decision
At least ¼" of effective rain <i>and</i> rain covers most (80%+) of demand	Skip - today's automatic starts are held.
Rain covers part of demand	Reduce - runs are shortened proportionally, never below 50% of programmed time.
Little or no rain, or demand is negligible (dormant season)	None - schedules run exactly as programmed.

The status card explains today's decision in plain language, including the numbers behind it ("Rain: 0.42 in • Demand: 0.18 in evapotranspiration"), and Home carries a matching note whenever an adjustment was applied.

Location

By default the forecast location is detected from your internet connection. If it looks wrong, search a city name or postal code and save it as a **custom location** - the forecast and all adjustment decisions follow it.

What it never touches

- Manual watering - per-zone or a hand-started schedule - always runs as set.

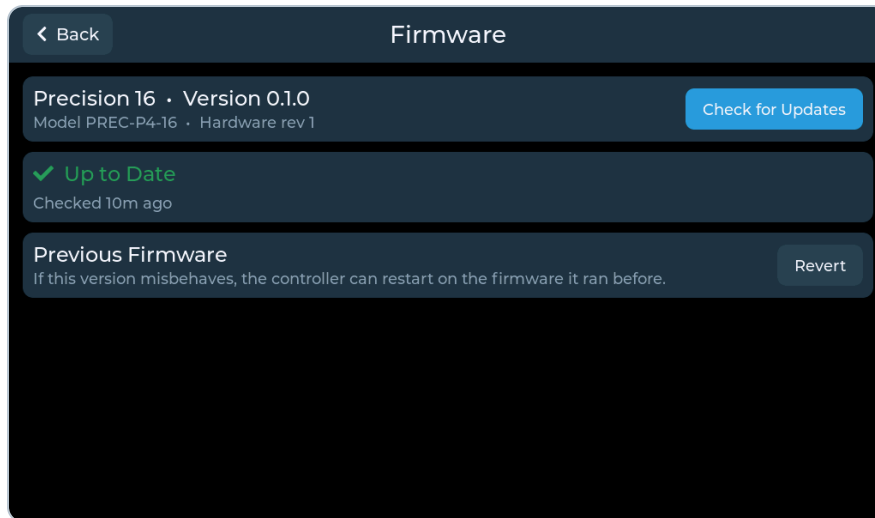
- When the toggle is off, or fresh forecast data is unavailable (Wi-Fi down, stale data, clock not synced), no adjustment is made and schedules run as programmed.

12. Frost Hold

Frost Hold is automatic freeze protection - there is nothing to configure. Whenever today's forecast low is **40°F or below**, automatic schedule starts are held and Home shows a Frost Hold banner with the forecast low. When the forecast warms, schedules resume on their own.

- **Manual watering still works** during a frost hold - deliberately, so you can blow out lines when winterizing the system.
- Like every weather feature, Frost Hold fails open: with no current forecast available, the hold stands down rather than blocking irrigation on stale data.

13. Firmware Updates



The Firmware screen (`Settings > Firmware`) keeps your controller current with free updates delivered over Wi-Fi. The identity card shows the product, installed version, and hardware revision.

Updating

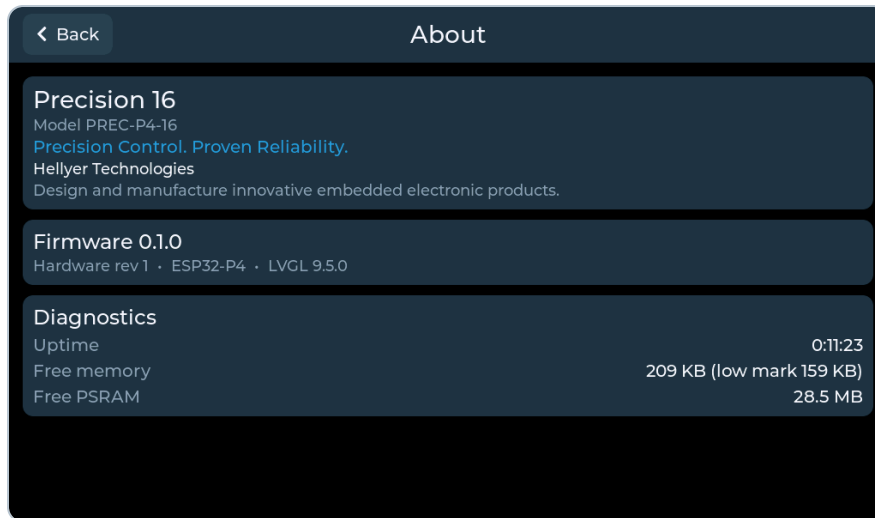
1. The controller checks for updates automatically once a day; **Check for Updates** checks on demand.
2. When an update is available, the Settings row and this screen call it out. Tap **Install Update** to download - a progress bar tracks the transfer, and watering is unaffected while it runs.
3. Once the download is verified, tap **Restart Now**. The screen goes dark for a few seconds while the update installs, then the controller starts up on the new version. Install actions wait until no zones are running.

Note: Leaving the Firmware screen after a download asks whether to discard the downloaded update - it stays available to download again. A power loss before or during the restart is safe: the controller simply starts on its current firmware with the update still offered.

Built-in safety

- Every download is verified against its published checksum before it touches the controller's firmware storage.
- A freshly installed update runs on probation for its first boots - if it fails to start reliably, the controller automatically rolls back to the previous firmware.
- **Revert** - after a successful update, a recovery card offers a manual return to the previous version if you ever need it.

14. About



The About screen (`Settings > About`) identifies your controller and shows its live health:

- **Product** - Precision 16, model PREC-P4-16, from Hellyer Technologies.
- **Firmware & hardware** - installed firmware version, hardware revision, processor, and interface library version. Useful when contacting support.
- **Diagnostics** - live uptime and free-memory readouts, updated every second.

15. Troubleshooting

Symptom	Things to check
A schedule didn't run	Is the schedule enabled? Is Standby Mode on (Home shows a banner)? Was it delayed from the Schedules screen? Did Weather Adjustment skip it (Home shows a rain note) or Frost Hold pause it (banner)? Has the clock synced?
Clock shows "--:--"	The controller is waiting for network time. Check Wi-Fi is connected; on restrictive networks, set a custom time server under Date & Time.
Wi-Fi is connected, but weather, clock, or updates do not work	A Wi-Fi connection is not the same as internet access. Verify the network itself can reach the internet - for example, open a website on a phone connected to the same Wi-Fi. Guest networks and captive portals (networks that show a login page) often block devices like the controller.
Wrong weather location	Set a custom city or postal code on the Weather Adjustment screen - the forecast follows it.
Watering seems shorter than programmed	Weather Adjustment reduced it after rain - the Home note and the Weather Adjustment screen show today's decision and the numbers behind it. Turn the feature off to always run full times.
Screen is dark	The display is asleep - touch it to wake. Adjust Display Sleep in Settings. The display turns on by itself whenever watering starts, so a dark screen also means nothing is currently running.
Nothing waters, everything looks right	Check Standby Mode is off, then run one zone from Manual Watering: if it runs, scheduling is the issue; if not, check valve wiring and water supply.
Update won't install	Installs wait while zones are running or queued. Stop watering (or wait for it to finish), then try again from the Firmware screen.